Hindi Fluency Analysis in VFT and SpAM: An IEEE-Style Report

Student Project Draft
BRSM course project

TABLE I
DATASET SUMMARY USED IN THE REPORT

Item	Value
Raw rows in CSV	1040
Hindi/Hinglish rows used in analysis	723
Participants	35
Domains	4
Animals responses	233
Foods responses	258
Colours responses	41
Body-parts responses	191

***Abstract*—This report studies Hindi/Hinglish verbal fluency data with a mix of descriptive statistics, mixed-effects models, nonparametric tests, semantic clustering, and SpAM-based alignment checks. The data cover four domains: animals, foods, colours, and body-parts. The main pattern is clear: response times are right-skewed, within-cluster retrieval is much faster than between-cluster retrieval, and semantic structure is stronger than phonological structure in several parts of the analysis. At the same time, some expected effects are weak or not stable, so the final picture is mixed, not perfect.**

***Index Terms*—verbal fluency, Hindi, Hinglish, SpAM, clustering, reaction time, semantic similarity, mixed-effects model**

I. INTRODUCTION

Verbal fluency tasks ask a participant to name as many words as possible from a category in a short time. This type of task is often used to study lexical access, semantic organization, and executive control [2], [1]. In this project, the response data are in Hindi/Hinglish and they are paired with SpAM coordinates, so the analysis can look at both response speed and spatial organization.

SpAM gives each word a position in a two-dimensional space. Words that are close in meaning should also be close in space [3]. In this dataset, the four domains are animals, foods, colours, and body-parts. The goal is to see whether the retrieval pattern is simple or whether it changes across domains, positions, and clustering rules.

II. DATA AND BACKGROUND

The main file is `merged_vft_spam_responses_enriched.csv`. The analysis uses the Hindi/Hinglish subset after a basic reaction-time filter of 60 seconds. The final analytic sample has 723 rows from 35 participants. The low-N colours domain is kept for description, but it is excluded from some inferential tests because it has only 4 participants.

The BRSM syllabus for the course also matches this kind of work well, because it focuses on descriptive statistics, hypothesis testing, ANOVA, correlation, regression, and multivariate methods [1]. That is why the report follows a standard statistics structure: first description, then method, then tests, then conclusion.

III. METHODS

A. Preprocessing

The notebook loads the enriched CSV, checks the required columns, and keeps rows with `rt_ms <= 60000`. It then splits the data into Hindi/Hinglish and English rows. The Hindi/Hinglish part is the main sample for this report.

The analysis uses response time in milliseconds (IRT), word position, subject ID, domain, SpAM coordinates `x` and `y`, and the confidence variable `hi_confidence`. For the semantic work, the notebook also uses multilingual sentence embeddings from `paraphrase-multilingual-MiniLM-L12-v2` [4].

B. Descriptive analysis

First, the report checks the overall shape of IRT values with histogram, KDE, and boxplots. It also calculates mean, median, skewness, and Shapiro-Wilk diagnostics by domain. This gives the basic distribution picture before any hypothesis test.

C. Cluster and position analysis

For the semantic clustering hypothesis, the notebook builds a simple cluster rule using an adaptive threshold of mean IRT plus one standard deviation within each subject-domain sequence. It then compares within-cluster IRT and between-cluster IRT with a Welch t-test and Cohen's d.

For the lexical exhaustion test, the notebook fits mixed-effects models of the form rt_ms $\sim$ position with subject-level random effects. This is used to check whether later positions become slower.

D. Domain tests

Domain differences in productivity are tested with Kruskal-Wallis on participant-level word counts. Domain differences in IRT are also tested with Kruskal-Wallis on response-level IRTs. Pairwise Mann-Whitney U tests are used as follow-up checks when needed [6].

E. SpAM, clustering, and composite score

For the SpAM part, the notebook compares semantic and phonological structure with adjusted rand index (ARI), sign tests, and a permutation-style check. HDBSCAN was not available in the environment, so agglomerative clustering was used as a fallback for one of the clustering comparisons.

The later analysis checks whether SpAM and semantic distances move together, whether neighbourhood density predicts retrieval speed, and whether phonological similarity grows later in the retrieval order. Finally, the notebook builds a composite fluency score from total words, speed, spatial compactness, semantic alignment, and phonological alignment, then compares it with a VFT-only score.

IV. Results

A. Overall pattern

The IRT distribution is clearly right-skewed. The histogram has a long tail, so the mean is above the median. The boxplot also shows large spread across the four domains. This is normal for fluency data, where some words come quickly and some words take much longer.

B. Within-cluster vs between-cluster retrieval

H1 is supported. Between-cluster retrieval is much slower than within-cluster retrieval. Overall, within-cluster IRT is 5841.9 ms and between-cluster IRT is 10960.3 ms. The Welch test gives $t = 9.2995$, one-tailed $p < 0.001$, and Cohen's $d = 1.1206$. This is a large effect.

The same pattern appears in every domain. Animals, foods, colours, and body-parts all show faster within-cluster retrieval than between-cluster retrieval.

C. Serial position and lexical exhaustion

H2 is not supported. The slope is mixed by domain. Foods and body-parts have negative slopes, animals is also negative but weaker, and colours is positive but not significant. The model support count is 0 out of 4 domains. So there is no clear and stable lexical-exhaustion pattern in this sample.

D. Domain differences in productivity and speed

The domain-level Kruskal-Wallis tests are not significant. For productivity, $H = 3.2639$, $p = 0.1956$. For IRT, $H = 3.9028$, $p = 0.1421$. This means that, once the low-N rule is applied, the three inferential domains do not separate strongly on these two measures.

E. Semantic and SpAM structure

H5 gives a positive semantic-SpAM result. The average ARI is 0.1765 and the sign test is significant. The permutation estimate is 0.5022, so this effect is not very strong under the more conservative check, but the direction is still positive.

H6 is weaker. The average phonetic-SpAM ARI is 0.0745 and the sign test is not significant at 0.05. So phonological alignment is much less stable than semantic alignment.

At the word level, SpAM-FastText alignment is domain-specific. Colours shows a small positive correlation ($\rho = 0.139, p = 0.0093$), while animals, foods, and body-parts do not show the same clear pattern. This means that semantic structure exists, but it is not equally strong in every category.

F. Phonological and neighbourhood effects

H7 is not supported. The interaction between similarity type and position is negative and not significant ($\beta = -0.004144, p = 0.1028$). The semantic slope and phonological slope are both small and not significant.

H8 is also not supported. Neighbourhood density has only a weak relationship with retrieval speed ($\rho = 0.114, p = 0.2570$). The data do not show a reliable speed benefit from denser neighbourhood structure.

G. Composite fluency score

H9 does not show an advantage for the integrated score. The integrated score and the VFT-only score have the same Spearman correlation with Hindi confidence ($\rho = -0.461, p = 0.0053$). The Steiger test gives $p = 1.000$, so there is no significant difference between them. In simple terms, the extra SpAM-based features do not improve this confidence link in the current sample.

H. Clustering agreement

The clustering methods agree quite well in most domains. K-means vs agglomerative ARI is 0.905 for animals, 0.861 for foods, 0.714 for colours, and 0.461 for body-parts. This suggests that the main semantic structure is not an accident of one single clustering method. The body-parts domain is the weakest one.

V. Conclusion

The report shows that Hindi/Hinglish fluency is not random. The strongest result is the big gap between within-cluster and between-cluster retrieval. Semantic structure is also visible in the clustering and SpAM work, but the effect is not equally strong in every domain. On the other hand, serial-position exhaustion, domain differences in productivity, neighbourhood-density speed effects, and the composite-score improvement are not stable in this sample.

For the next step, I would improve the dataset with a larger colours sample, add a cleaner manual coding pass for semantic clusters, and compare the current notebook results with a human-rated poster version. I would also replace the placeholder citations with the final bibliography.

References

[1] BRSM course syllabus and topic summary, project handout used for this report. Placeholder entry to be replaced with the final course reference.

[2] General background on verbal fluency tasks and lexical retrieval. Placeholder entry to be replaced with the final source.

[3] General background on the Spatial Arrangement Method (SpAM). Placeholder entry to be replaced with the final source.

[4] General background on multilingual sentence-transformer embeddings used for semantic analysis. Placeholder entry to be replaced with the final source.

[5] General background on mixed-effects models for repeated-measures data. Placeholder entry to be replaced with the final source.

[6] General background on Kruskal-Wallis and Mann-Whitney U tests for nonparametric comparisons. Placeholder entry to be replaced with the final source.

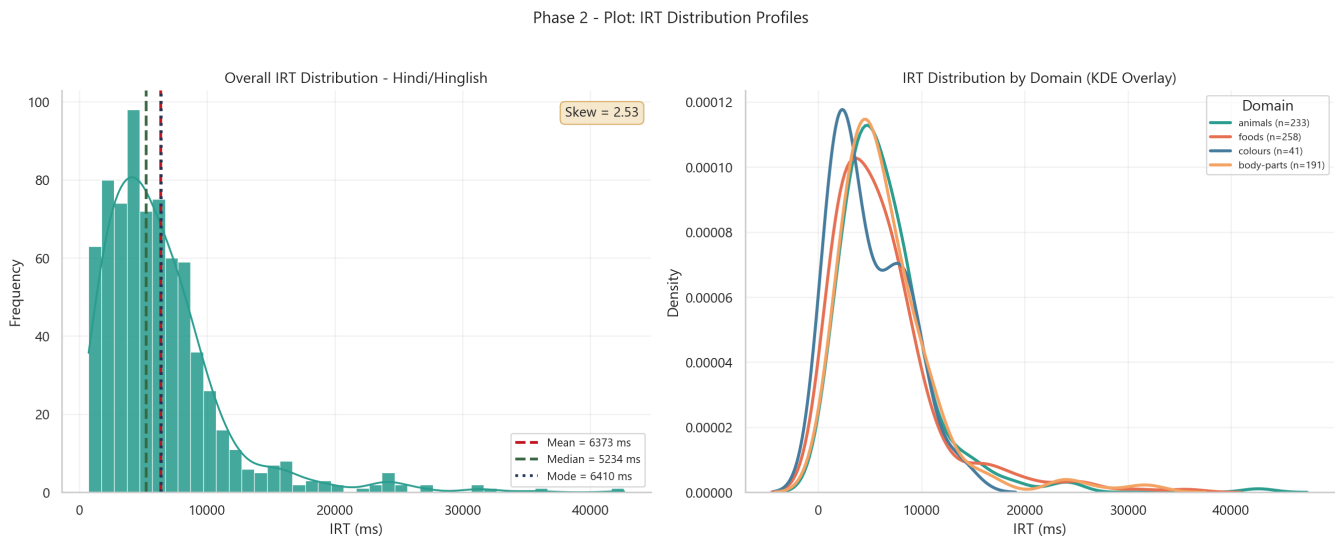

Fig. 1. Overall IRT distribution with KDE and domain overlays. The distribution is right-skewed and has a long tail.

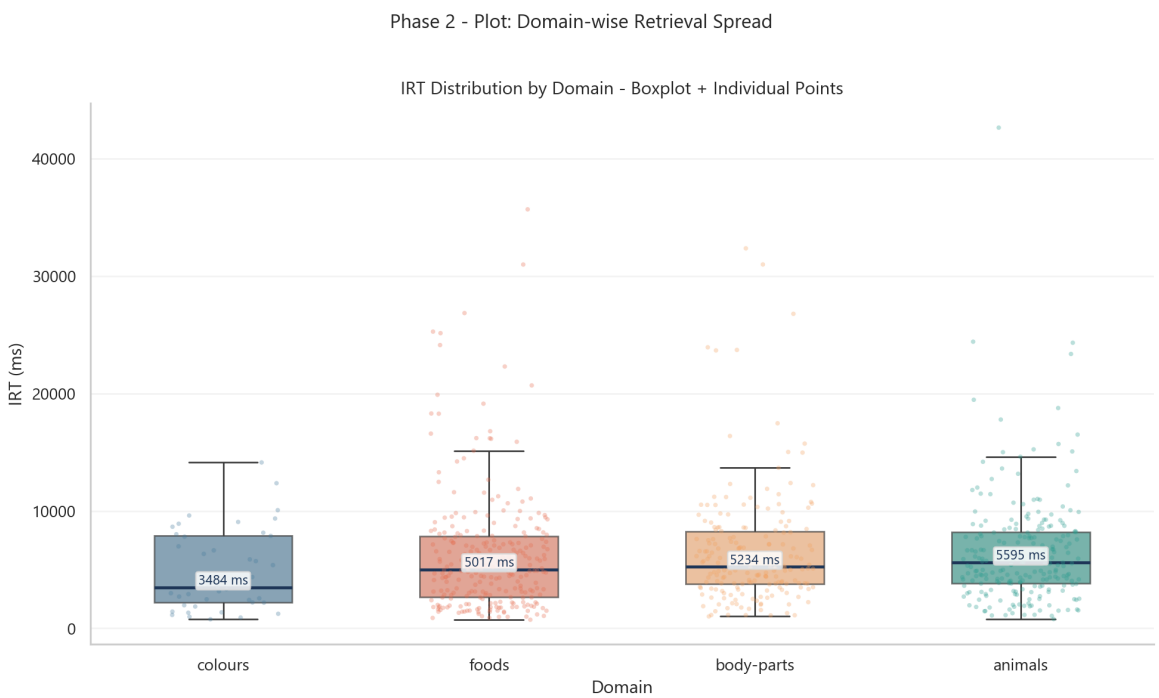

Fig. 2. IRT by domain. Colours is the smallest group and is mainly used for description, not for the main inferential tests.

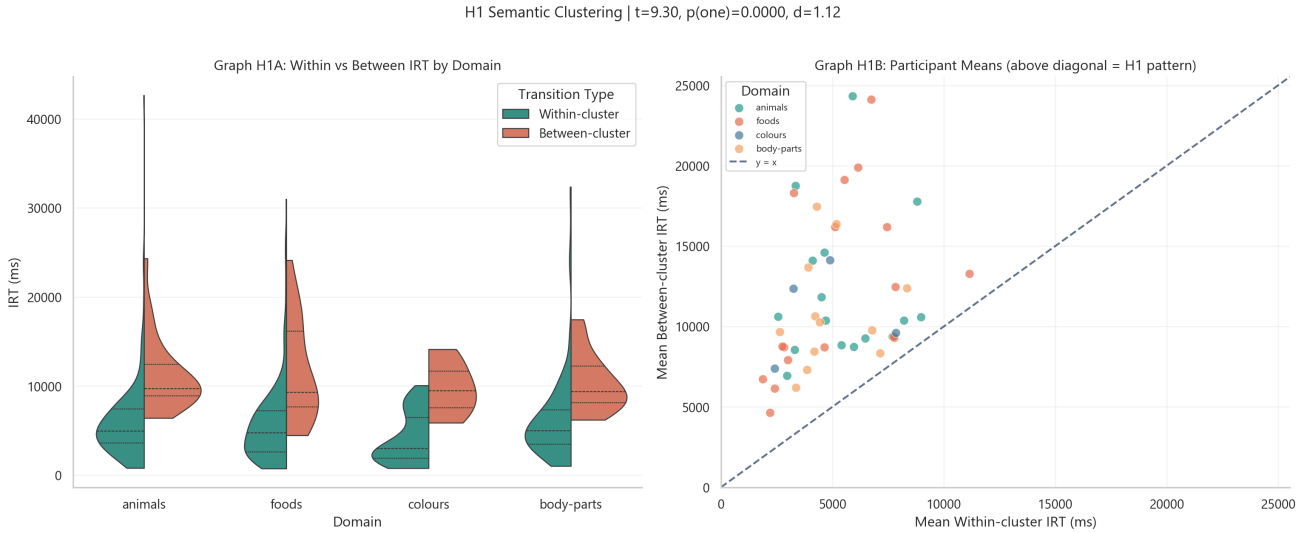

Fig. 3. Within-cluster and between-cluster IRT comparison for H1. The between-cluster side is consistently slower.

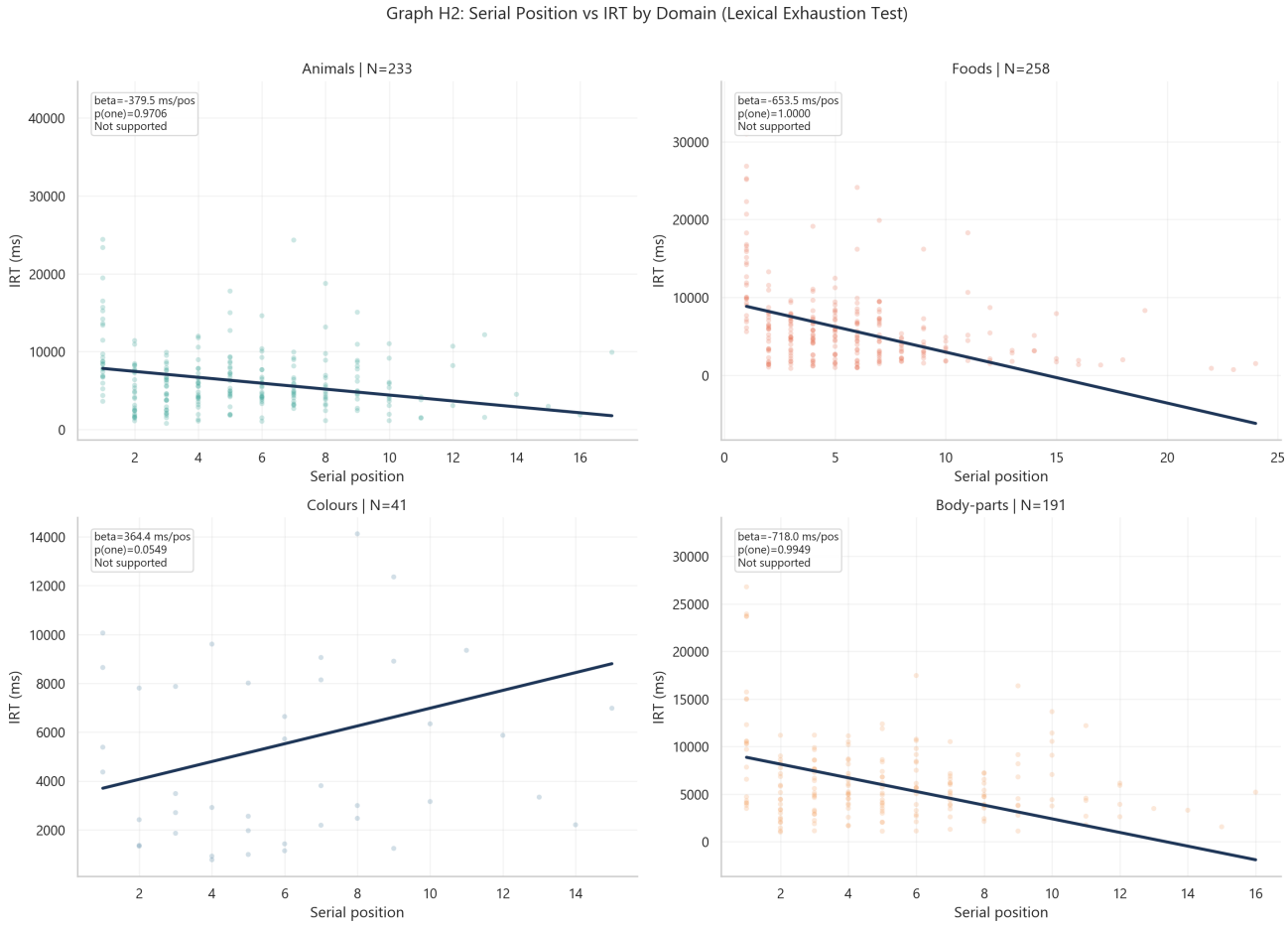

Fig. 4. Serial position mixed-effects models for H2. The slopes do not show a stable lexical-exhaustion pattern.

TABLE II
KEY HYPOTHESIS RESULTS FROM THE NOTEBOOK

Hyp.	Main statistic	p-value	Decision	Short reading
H1	Welch $t = 9.2995$, $d = 1.1206$	< 0.001	Supported	Between-cluster IRT is much slower.
H2	Mixed-effects slope on position	Mixed	Not supported	No stable lexical-exhaustion trend.
H3	Kruskal-Wallis $H = 3.2639$	0.1956	Not supported	Productivity is similar across inferential domains.
H4	Kruskal-Wallis $H = 3.9028$	0.1421	Not supported	IRT is similar across inferential domains.
H5	Mean ARI = 0.1765; sign test positive	< 0.001	Partly supported	Semantic-SpAM alignment is clearly positive.
H6	Mean ARI = 0.0745	0.0662	Not supported	Phonetic-SpAM alignment is weak.
H7	Interaction $\beta = -0.004144$	0.1028	Not supported	No later phonological gain.
H8	Spearman $\rho = 0.114$	0.2570	Not supported	Neighbourhood density does not predict speed.
H9	$\rho = -0.461$ for both scores; Steiger $p = 1.000$	1.0000	Not supported	Integrated score does not beat VFT-only.

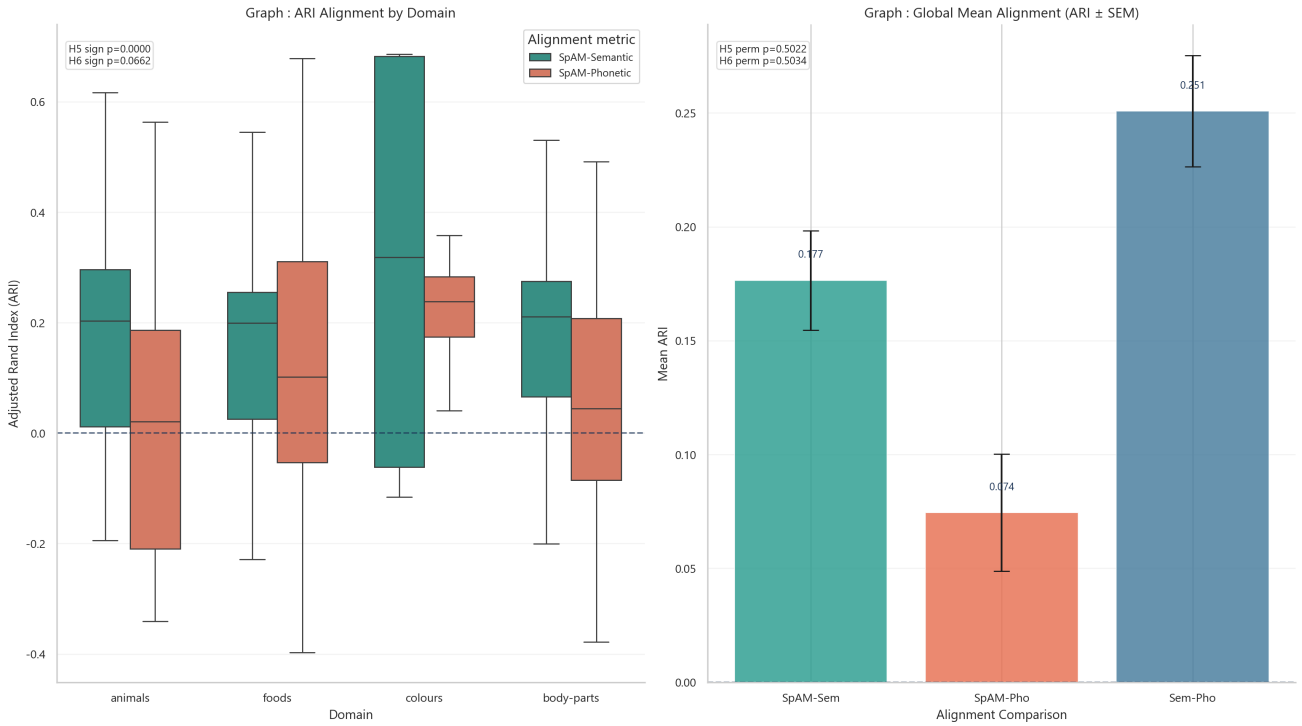

Fig. 5. H5 and H6 alignment metrics. Semantic alignment is stronger than phonetic alignment.

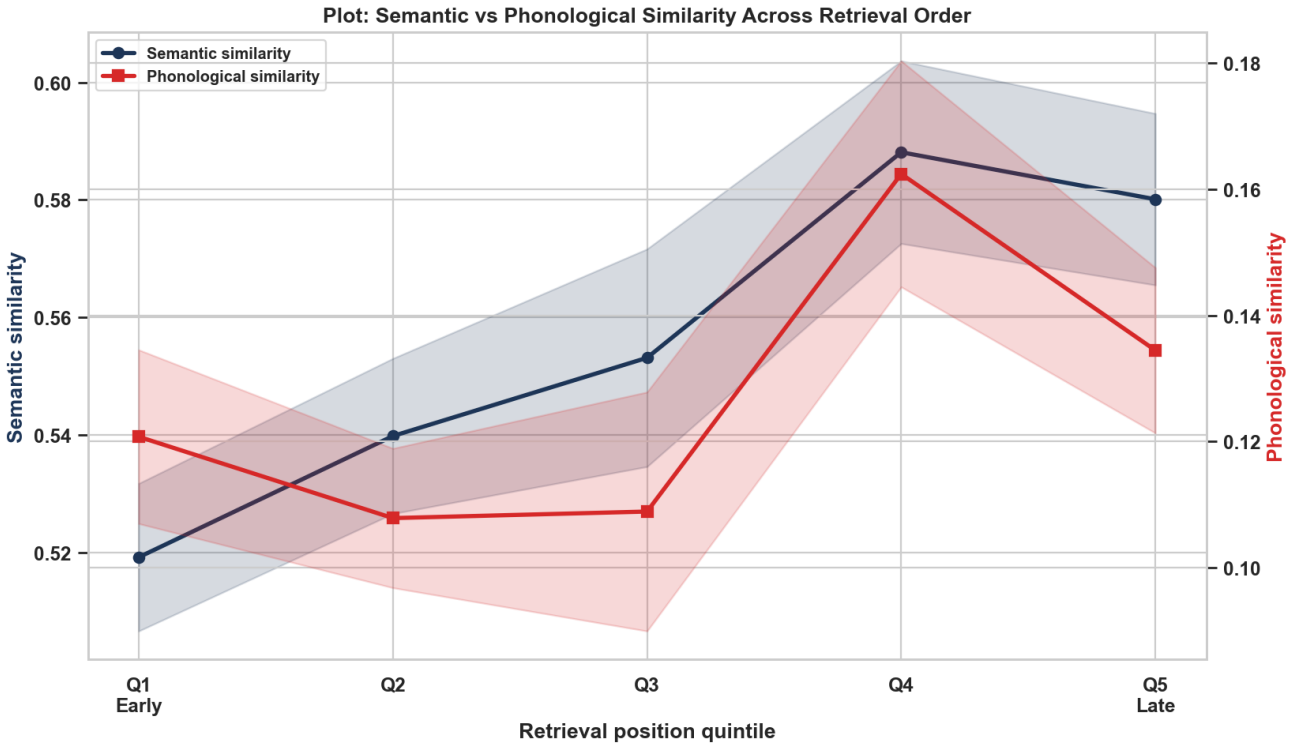

Fig. 6. H7 dual-similarity check. Phonological similarity does not overtake semantic similarity in a reliable way.

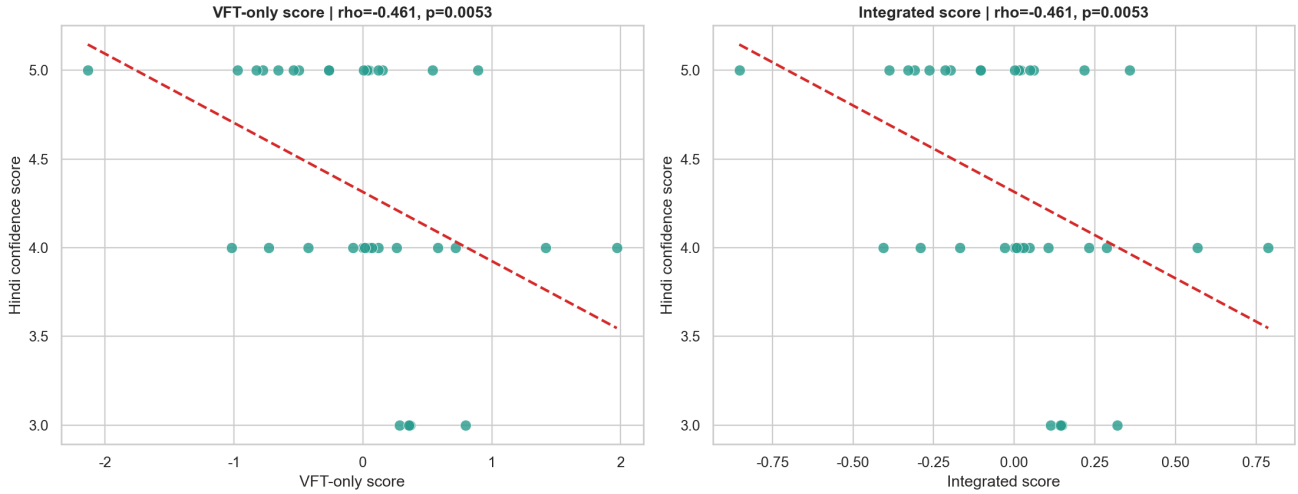

Fig. 7. Composite score versus Hindi confidence. The integrated score does not outperform the VFT-only score.

TABLE III
SpAM AND SEMANTIC STRUCTURE BY DOMAIN

Domain	KM-AG ARI	SpAM-FastText ρ	Reading
Animals	0.905	-0.070	Weak alignment
Foods	0.861	-0.036	Weak alignment
Colours	0.714	0.139	Best alignment
Body-parts	0.461	-0.033	Weak alignment